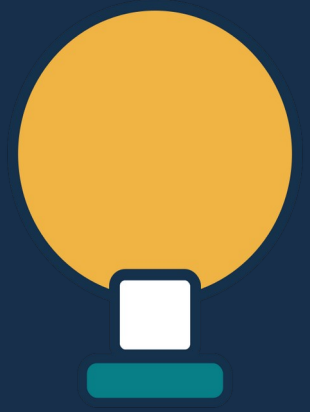
ESSENTIAL LECTURE 06

Introduction to Vector Graphics



Draw, arrange and export one simple educational icon

Shapes • Fill and stroke • Alignment • SVG / PNG

What you should be able to do

This lecture concentrates on a small number of useful vector-design skills.

EXPLAIN

Describe vector graphics and identify when vector artwork is more suitable than a raster image.

CREATE

Use basic shape tools, fills, strokes and alignment to draw a simple educational icon.

DELIVER

Save an editable source file and export a usable SVG or PNG for a publication.

One well-made, editable icon is better than several unfinished designs.

From photographs to illustrations

Think back to the images and poster developed in Lectures 4 and 5.

Raster work

Photographs are usually raster images made from pixels. Cropping and resizing can change pixel detail, so enlargement needs care.

Vector work

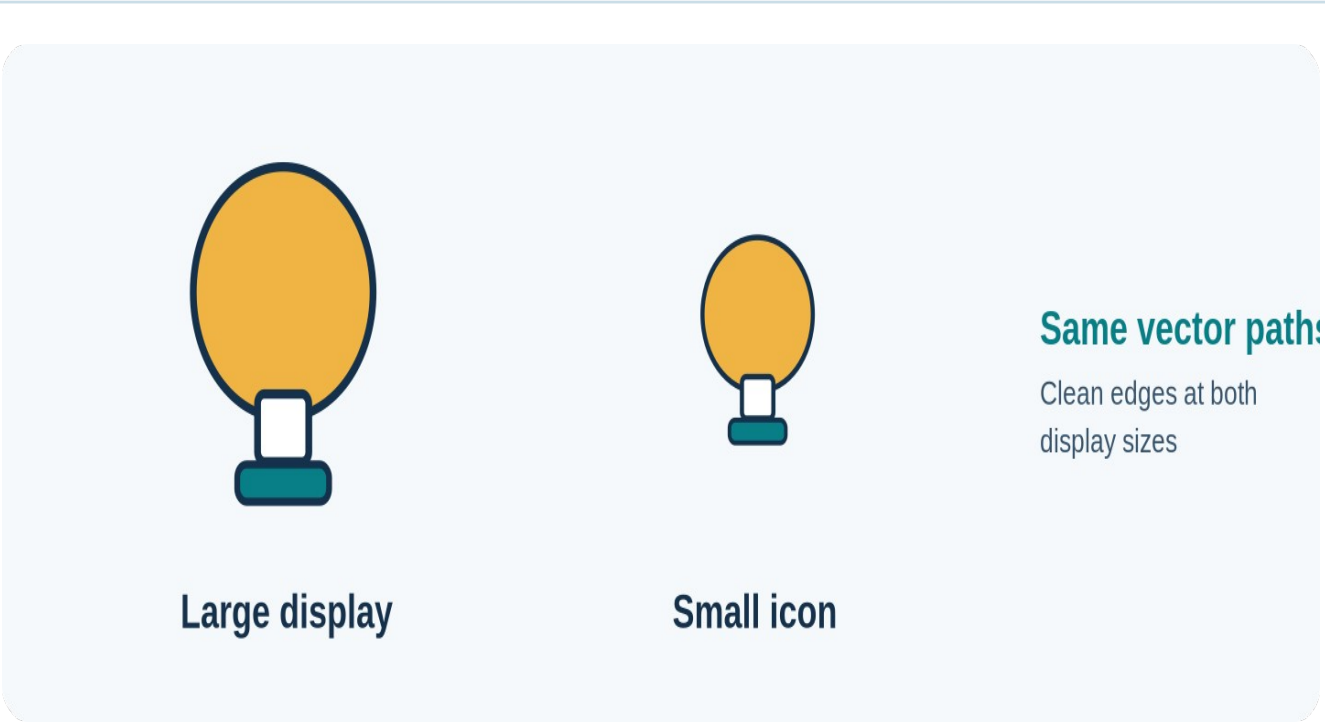
Simple symbols are easier to make with vector paths. Their outlines can be edited and scaled without depending on a fixed pixel grid.

Different tools solve different publishing problems.

KEY CONCEPT

What is a vector graphic?

A vector graphic is constructed from mathematically described shapes and paths.



Main advantage

Vector shapes can be enlarged or reduced while preserving clean outlines.

Where useful

Icons, logos, diagrams, signs and other illustrations built from simple shapes.

COMPARISON

Vectors and pixels are not the same

Both formats are useful; choose according to the material being produced.

Raster graphic

Contains a grid of pixels. It suits photographs and detailed textures. Enlarging a small raster picture can expose pixelation.

Vector graphic

Contains paths, shapes, fills and strokes. It suits icons and line illustrations that must remain crisp at different sizes.

For a portrait photo, keep the raster image; for a simple icon, consider vector artwork.

Why use vector graphics in publishing?

Students will encounter the same symbol on documents of different sizes.

Scale

The same artwork can appear as a small worksheet icon or a large classroom poster without rebuilding its shapes.

Edit

Individual objects, colours and outlines can be changed independently.

Reuse

The symbol can be placed in brochures, hand-outs, slides and online announcements.

What do Illustrator and Inkscape do?

Both applications allow objects to be edited as vector artwork.

Adobe Illustrator

A commercial vector-design program used to create and edit logos, diagrams and illustrations. In this lesson we need only its shape, fill, alignment and export tools.

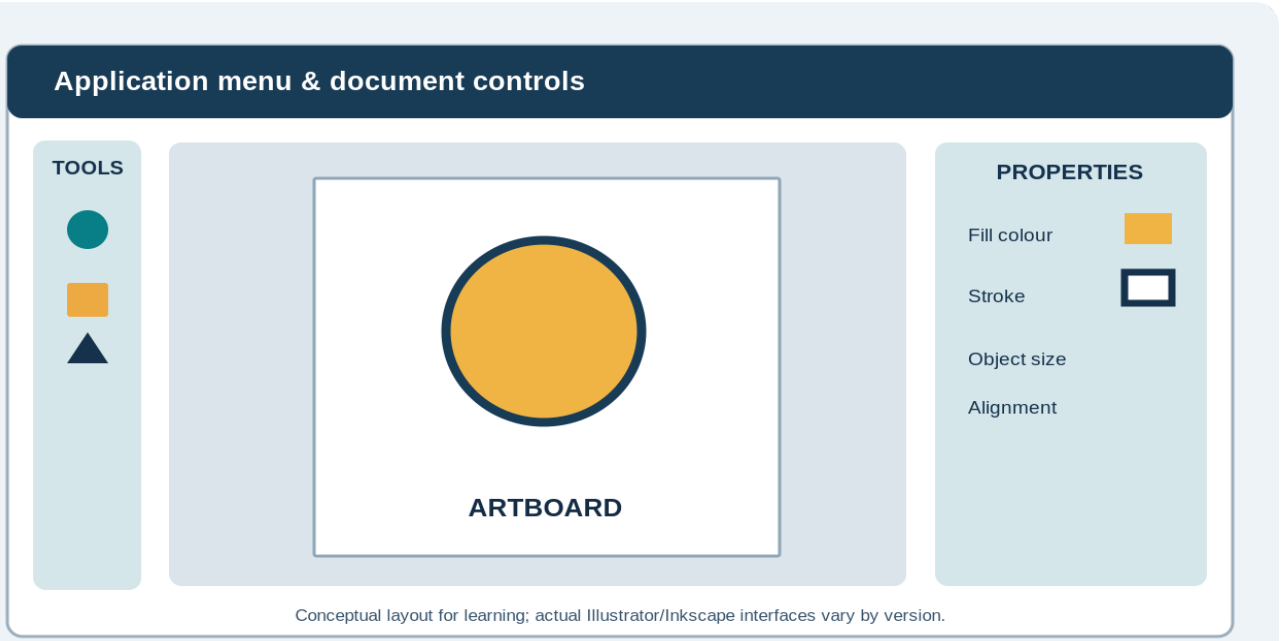
Inkscape

A free vector-editing alternative. Its buttons and menus differ, but the essential ideas of objects, fills, strokes, grouping and export are similar.

Learn the design ideas first; the particular software version may vary.

Understanding the vector-editing workspace

This is a conceptual diagram, not a screenshot of a specific application version.



Tools & artboard

The shape tools create objects. The artboard is the page or area prepared for export.

Properties

Fill, stroke and object size can usually be changed through toolbars or panels.

What is an artboard?

The artboard defines the intended output area, not the mathematical sharpness of a vector path.

Example setup

For practice, create an 800 × 800 unit square artboard. In Illustrator, choose File > New and enter the dimensions. Set a clear document name.

Important distinction

A vector object is resolution-independent while kept as a vector. When exported as a PNG, it becomes a pixel image with a selected width and height.

Keep the original vector file even after producing PNG copies.

Three basic shape tools

You need very few tools to build the required icon.

Ellipse Tool

Creates circles and ovals. Hold the appropriate modifier key to constrain proportions if your software supports it.

Rectangle Tool

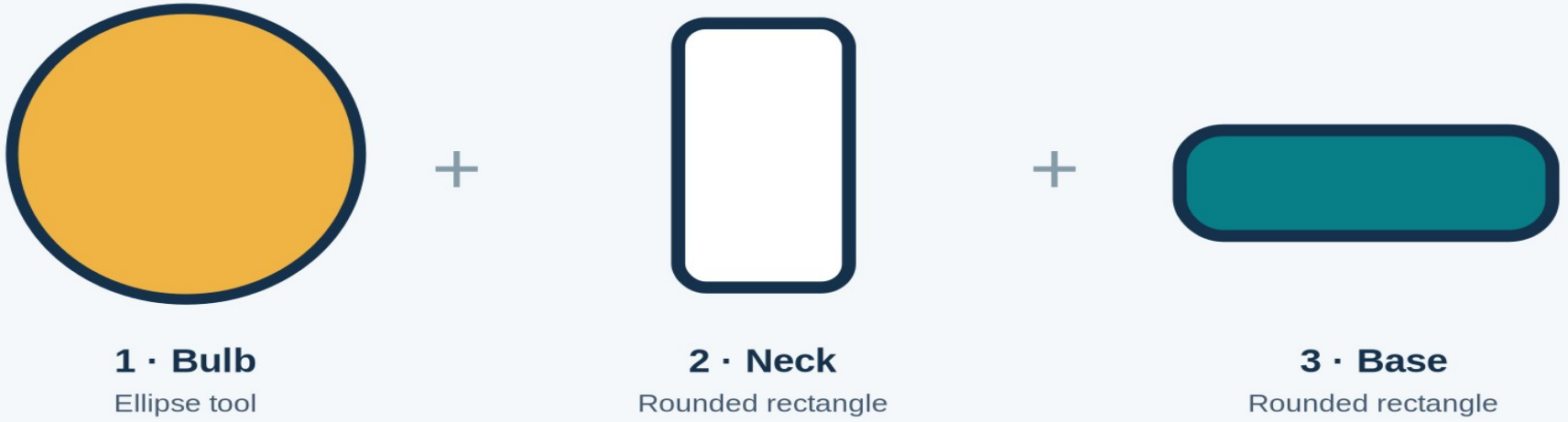
Creates rectangular elements, such as banners, simple pages and icon bases.

Selection Tool

Moves, scales and selects the shapes already drawn. Use it before changing the objects' positions.

Our three-shape learning icon

A recognisable symbol can be built without complicated path drawing.



Create each part as a separate object first; then align and group them.

Two properties give shapes their appearance

Fill and stroke are separate controls in a vector editor.

Fill

The fill is the interior colour of a shape. In our model icon, a warm gold fill identifies the main bulb. A shape can also be set to no fill.

Stroke

The stroke is the line along the shape’s boundary. A dark outline can improve the symbol’s definition, especially when used against a pale page.

Use only a small number of colours and keep the outlines consistent.

Fill and stroke: practical decisions

Avoid choosing colours just because they are available.

Use colour with a purpose

The bulb can be gold to suggest an idea; the base can be teal to fit the Tumaini project. A limited palette makes the symbol look intentional.

Check readability

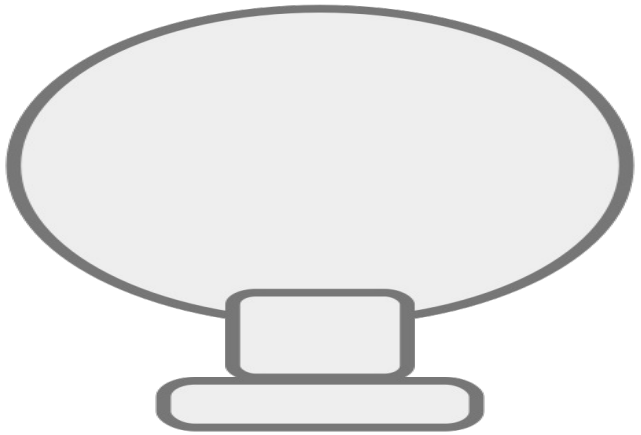
A very thin outline may disappear at small sizes. If the icon looks cluttered, simplify its details rather than adding more colour.

A strong simple icon still works when it is reduced in size.

WORKED EXAMPLE

Construct the lightbulb from three shapes

Start by drawing each object at approximately the correct size.



1 — Bulb

Draw an ellipse at the upper centre of the artboard.

2 and 3 — Neck/base

Draw two rounded rectangles beneath it; position them using the same vertical centre.

Layer order and shape overlap

The position of objects on top of one another can affect the finished icon.

In front / behind

A new shape may cover part of an older one. Use Arrange or the object-order controls to bring an element forward or send it behind.

Our icon

The narrow neck should meet the ellipse cleanly, and the base should connect beneath it. Small overlaps are acceptable if no unwanted lines remain visible.

Check overlap at normal size rather than only while zoomed in closely.

Align shapes to a shared centre

Good alignment makes a simple icon look deliberate.

Before alignment

A bulb, neck and base may be slightly off-centre when drawn with the mouse. The error can make the icon appear unstable.

After alignment

Select the objects and use horizontal-centre alignment. If your application aligns to the whole selection, check whether the result matches the intended visual centre.

Do not rely only on guesswork; use alignment tools and then inspect visually.

Select, move and resize objects

These are everyday operations in vector design.

Selection

Use the Selection Tool to click one shape or select several objects. The bounding box shows which objects are active.

Resizing

Drag a handle while preserving proportions when needed. If the bulb becomes too narrow or too wide, the icon may lose its intended meaning.

Deselect and examine the full composition after every major adjustment.

Why group related objects?

Grouping makes it easier to move a finished icon without losing its construction.

Before grouping

Each part is independent. Moving only the bulb could leave the neck and base behind.

After grouping

Select all three shapes and choose Group (commonly Ctrl+G). They can now move together while remaining editable when you enter or ungroup the set.

Save a grouped version for reuse, but keep the vector objects editable.

Three tests for a useful icon

Apply these checks before you export the design.

Recognisable

Can someone identify the idea represented without an explanation? If not, simplify or adjust the proportions.

Consistent

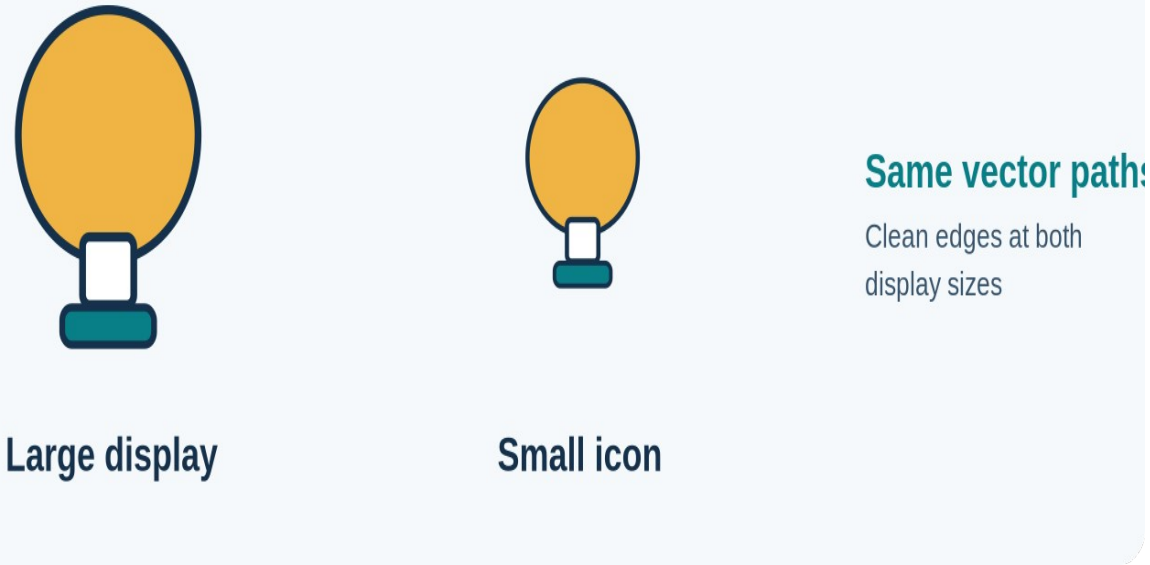
Are the edges, colours, alignment and shape relationships tidy?

Reusable

Will the icon remain clear against a white or coloured background and when reduced?

Inspect the icon at more than one size

A design that looks good when large can still fail as a small publication symbol.



Large view

Inspect the overall silhouette and the relative size of each shape.

Small view

Check that the neck and base remain visible and no important detail has become too thin.

The editable source and exported copy have different purposes

Do not submit only a flattened picture if the task asks for editable vector artwork.

Source file

Illustrator normally uses .ai as its native source format. Inkscape uses SVG naturally. The source retains editable vector objects.

Exported copy

SVG retains vector paths for compatible uses. PNG is a raster image and can preserve transparency; its sharpness depends on its exported pixel dimensions.

Submit a source file AND one suitable export.

When should you use SVG or PNG?

Choose the format based on where the image will be used.

SVG

Good for simple icons, web graphics and other situations where scalable vector content is supported. Check compatibility with the destination software.

PNG

Useful when an application needs a widely supported raster picture or a fixed-size image. Select sufficient pixels for the intended output.

Neither format is automatically best for every printing or publishing workflow.

How to save the Illustrator source

Save early and use a descriptive file name.

Illustrator route

Choose File > Save As or Save, then save an Illustrator (.ai) document to your project folder. Preserve the shapes as editable objects.

Inkscape route

Save the design as SVG. Open the saved file again to confirm that objects can still be selected and recoloured.

Avoid names such as “final2newlatest” that make versions difficult to identify.

How to export the icon

Export only after inspecting the source at normal and small size.

SVG export

Choose a suitable SVG save/export route. Reopen the SVG to confirm that all three objects appear and the colours remain correct.

PNG export

Use the export controls to select the artboard or drawing, then choose the pixel dimensions. Inspect the output against a pale and a dark background.

Exporting a picture does not replace the editable source file.

Practical: build the lightbulb icon

Follow this sequence using Illustrator or Inkscape.

1. Create an 800 × 800 square document and give it a sensible file name.
2. Draw one ellipse and two rounded rectangles using shape tools.
3. Apply a small colour palette and a consistent outline.
4. Centre the objects and adjust their order until the icon reads clearly.

Work at a comfortable zoom, then check the whole icon at normal size.

Practical: finish and submit

Keep the final requirements modest and achievable.

1. Select the three objects and group them after checking their positions.
2. Save the editable vector source (.ai or .svg, depending on the editor).
3. Export a PNG copy and, where appropriate, an SVG copy.
4. Reopen the saved file and verify that the shapes remain editable.

The source and export should represent the same finished icon.

Place the icon on an educational hand-out

An icon should support a publication rather than compete with its message.



Purpose

A learning symbol helps readers recognise the publication’s educational theme.

Where to use it

Use a small icon beside a heading or in a clear information section, while leaving enough white space.

Common beginner mistakes

Most of these problems can be corrected without an advanced tool.

What goes wrong	What to do
Shapes are stretched, not centred, poorly overlapped or coloured inconsistently. Sometimes the icon is exported with an unwanted background.	Undo and resize proportionally, align shared centres, inspect overlaps, limit colours and check export settings. Keep an editable version before making large changes.

Always judge the appearance of the complete icon, not just individual shapes.

What if the icon needs revision next week?

Good file management saves time after the lesson.

Editable approach

Open the saved .ai or SVG file, select the component, change the fill or dimensions and export a revised copy.

Problematic approach

If only a small PNG remains, changing the individual shapes will be much harder. Enlarging that PNG may introduce pixelation.

Preserve source files with meaningful names in a clearly organised folder.

OPTIONAL EXTENSION

More advanced vector tools can wait

Begin with simple shapes, strokes, alignment and grouping before trying advanced vector tools.

If you finish early

Try one additional colour version or a monochrome version. Check whether the icon remains recognisable with only one colour.

For later study

Anchor points, Pen Tool, Pathfinder and complex shape construction can improve advanced artwork, but they are not required for this core practical.

Complete the essential three-shape exercise first.

How vector icons help different publications

Think about one symbol used in several places.

A4 hand-out

A modest icon can support a heading and establish a visual theme without distracting from the text.

Digital notice

A simple symbol can be placed beside a message and scaled to fit different digital layouts.

Poster

The same vector artwork can appear larger while retaining its crisp outline.

Explain your design choices

An important part of design is being able to justify what you have done.

Why these shapes?

The ellipse suggests the main bulb. The narrower neck and base make the silhouette resemble a lightbulb rather than a plain circle.

Why these colours?

Gold draws attention to the bulb while the teal base fits the learning-centre example. Readability remains more important than decoration.

There is no need for an elaborate logo story; explain the practical decisions clearly.

QUICK CHECK

Source versus export: one last comparison

Make sure you can explain these differences without looking at the notes.

Vector source (.ai / SVG)

Contains selectable paths or objects. Suitable for changing the icon’s colours, proportions and composition.

Raster output (PNG)

Contains pixels produced at the chosen export size. Easy to insert into many documents but less flexible for shape-by-shape editing.

A PNG export is a deliverable; the editable vector source is your working design.

What should be in your submission?

Use these checks before asking for feedback.

Source

An editable AI or SVG file with three clearly identifiable components.

Export

A PNG copy that displays cleanly and is not cropped unintentionally.

Explanation

Two or three sentences explaining the shapes, alignment and intended use of the icon.

Can you explain these key concepts?

Answer in sentences, not just isolated words.

Question 1

Explain how vector graphics differ from raster graphics and give an example of each.

Question 2

What is the difference between fill and stroke? How does alignment improve an icon?

Question 3

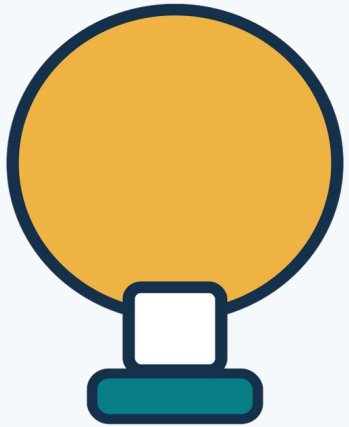
Why should a student keep an editable AI or SVG file after exporting a PNG?

A small icon, useful skills

The next step is to use the icon within a page layout.

What you have learned

A vector icon is made from editable shapes. You can create it with simple tools, align and group the pieces, save the source and export an appropriate copy. This is enough for our introductory exercise.



Next: use prepared images and icons in a readable A4 page layout with InDesign.